

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

August 2023

22MCA211 - DIGITAL IMAGE PROCESSING & PATTERN RECOGNITION

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a) Explain the concept of sampling and quantization.		10	L2	1	1
	b) Explain the following terms:					
	i) Neighbors of a pixel					
	ii) Distance function					
	iii) Euclidean distance					
	iv) City block distance		10	L2	1	1
2.	a) Explain Histogram Processing with at least two examples.		10	L2	1	1
	b) Explain the Process of Image Sensing and Acquisition.		10	L2	1	1
		Unit – II				
3.	a) Explain 2D-DFT. Explain the following properties of 2D-DFT:					
	i) Translation		10	L2	2	1
	ii) Rotation					
	b) Explain Image Smoothing and Image Sharpening Using Frequency Domain Filters.		10	L2	2	1
4.	a) Illustrate Restoration in the Presence of Noise Only using Spatial Filtering.		10	L2	2	1
	b) Explain the algorithm for frequency domain filtering with a block diagram.		10	L2	2	1
		Unit – III				
5.	a) Explain the general image compression model with a diagram.		10	L2	3	1
	b) Explain the Detection of Discontinuities and Edge Detection		10	L2	3	1
6.	a) Explain Erosion, Dilation with an example each.		10	L2	3	1
	b) Explain inverse filtering with an example.		10	L2	3	1
		Unit – IV				
7.	a) Illustrate Pattern recognition. Compare and contrast classification and clustering.		10	L2	4	1
	b) Explain the different distance functions and similarity measures.		10	L2	4	1
8.	a) Explain K-means Clustering algorithm.		10	L2	4	1
	b) Explain clustering methods and applications.		10	L2	4	1
		Unit – V				
9.	a) Explain the general framework for applying Bayes' decision rule with continuous features.		10	L2	5	1
	b) Illustrate Multilayer perceptron. Explain the key characteristics and applications of MLP.		10	L2	5	1
10.	a) Illustrate minimum distance classifier with an example.		10	L2	5	1
	b) Explain the following:					
	i) KNN Classifier					
	ii) Standardization and Normalization.		10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
